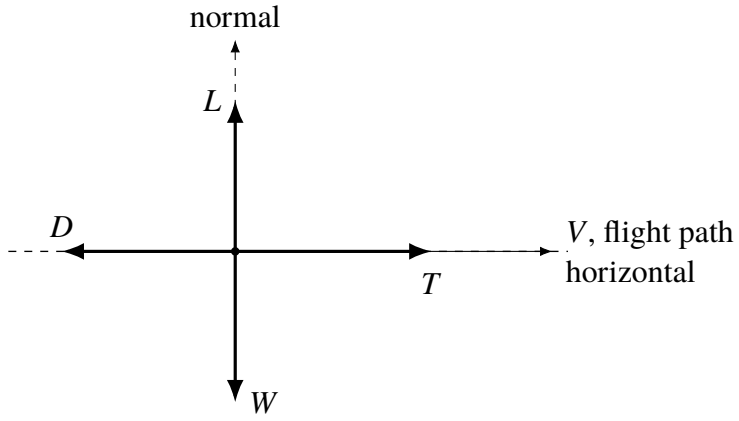
1 / STEADY LEVEL FLIGHT

Week 3: Equilibrium

Write and use the force and moment equations for **steady level flight, climb, descent and glide**.

Steady level flight: constant speed and height



V: airspeed; L: total lift; D: drag; T: thrust; $W = mg$: weight.

$$\text{Along the flight path:} \qquad T - D = 0,$$
$$\text{Normal to the flight path:} \qquad L - W = 0.$$

$$L = W, \qquad T = D.$$

Pitching equilibrium

$$M = 0.$$

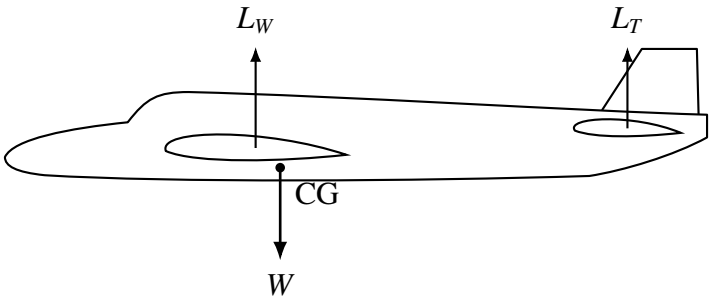
M is the net pitching moment. Nose-up is positive.

The aircraft is **trimmed** when its forces and pitching moment balance at the chosen flight condition.

1 / STEADY LEVEL FLIGHT

Total lift is shared by the wing and tail

Lift is the combined contribution of the wing–body and the tailplane. Take both loads as positive upwards in level flight.



1. Add the forces

$$L = L_W + L_T.$$

Using the same density ρ and speed V at both surfaces,

$$L = \tfrac{1}{2}\rho V^2 S_W C_L, \quad L_W = \tfrac{1}{2}\rho V^2 S_W C_L^W, \quad L_T = \tfrac{1}{2}\rho V^2 S_T C_L^T.$$

2. Divide by the whole-aircraft reference force

$$\frac{L}{\tfrac{1}{2}\rho V^2 S_W} = \frac{L_W}{\tfrac{1}{2}\rho V^2 S_W} + \frac{L_T}{\tfrac{1}{2}\rho V^2 S_T}\frac{S_T}{S_W}.$$

$$C_L = C_L^W + \frac{S_T}{S_W} C_L^T.$$

S_W and S_T are wing and tail planform areas. The area ratio appears because the two surface coefficients use different reference areas.

1 / STEADY LEVEL FLIGHT

Differentiate the lift equation at fixed controls

Geometry, tail setting and controls are fixed. The lift-response coefficients a_w, a_1, a_2, a_3 are constant, in $1/\text{rad}$. α and α_T are wing and tail incidence from their no-lift lines.

1. Differentiate the total lift relation

$$C_L = C_L^W + \frac{S_T}{S_W} C_L^T \quad \Longrightarrow \quad \frac{\partial C_L}{\partial \alpha} = \frac{\partial C_L^W}{\partial \alpha} + \frac{S_T}{S_W} \frac{\partial C_L^T}{\partial \alpha}.$$

2. Differentiate the wing contribution

a_w is the finite-wing lift slope called a in Week 2.

$$C_L^W = a_w \alpha \quad \Longrightarrow \quad \frac{\partial C_L^W}{\partial \alpha} = \frac{\partial (a_w \alpha)}{\partial \alpha} = a_w.$$

3. Differentiate the tail contribution

$$C_L^T = a_1 \alpha_T + a_2 \eta + a_3 \beta_{\text{tab}},$$

$$\frac{\partial C_L^T}{\partial \alpha} = a_1 \frac{\partial \alpha_T}{\partial \alpha} + a_2 \underbrace{\frac{\partial \eta}{\partial \alpha}}_{0} + a_3 \underbrace{\frac{\partial \beta_{\text{tab}}}{\partial \alpha}}_{0}.$$

The elevator angle η and tab angle β_{tab} are held fixed. From Week 2,

$$\alpha_T = \alpha + \eta_T - \varepsilon \quad \Longrightarrow \quad \frac{\partial \alpha_T}{\partial \alpha} = 1 + \underbrace{\frac{\partial \eta_T}{\partial \alpha}}_{0} - \frac{d\varepsilon}{d\alpha}.$$

The tail setting η_T is fixed; downwash ε changes with α. Hence

$$\frac{\partial C_L^T}{\partial \alpha} = a_1 \left(1 - \frac{d\varepsilon}{d\alpha}\right).$$

4. Substitute both contributions

$$\boxed{\frac{\partial C_L}{\partial \alpha} = a_w + \frac{S_T}{S_W} a_1 \left(1 - \frac{d\varepsilon}{d\alpha}\right).}$$

Neglecting the tail contribution gives $\partial C_L / \partial \alpha \simeq a_w$.

1 / STEADY LEVEL FLIGHT

Take moments about the wing lift application point

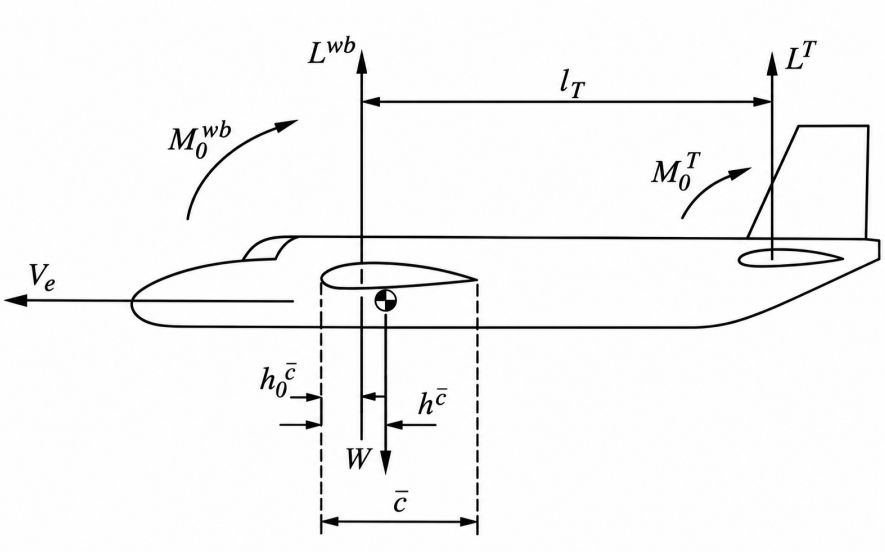


Figure notation: $L^{wb} = L_W$, $L^T = L_T$, $M_0^{wb} = M_0$, $V_e = V$. The separate tail couple M_0^T is neglected.

1. Read the positions and arms

$\bar{c}$ is the wing mean aerodynamic chord. The CG and wing AC lie $h\bar{c}$ and $h_0\bar{c}$ aft of its leading edge. Their separation is $(h - h_0)\bar{c}$. l_T is the arm from the wing aerodynamic centre (AC) to the tail AC.

2. Take moments about the wing lift application point

M is the net moment about the wing AC; nose-up is positive.

$$\begin{aligned} M &= M_0 + W(h\bar{c} - h_0\bar{c}) - L_T l_T \\ &= M_0 + W(h - h_0)\bar{c} - L_T l_T. \end{aligned}$$

Wing lift has zero arm about this point. Its aerodynamic couple M_0 remains.

3. Define the moment coefficient

$$C_M = \frac{M}{\frac{1}{2}\rho V^2 S_W \bar{c}}.$$

1 / STEADY LEVEL FLIGHT

Express the pitching balance in coefficients

1. Start from the same dimensional equation

$$M = M_0 + W(h - h_0)\bar{c} - L_T l_T.$$

2. Divide every term by $\frac{1}{2}\rho V^2 S_W \bar{c}$

$$\underbrace{\frac{M}{\frac{1}{2}\rho V^2 S_W \bar{c}}}_{C_M} = \underbrace{\frac{M_0}{\frac{1}{2}\rho V^2 S_W \bar{c}}}_{C_{M0}} + \frac{W(h - h_0)\bar{c}}{\frac{1}{2}\rho V^2 S_W \bar{c}} - \frac{L_T l_T}{\frac{1}{2}\rho V^2 S_W \bar{c}}.$$

Weight term: cancel $\bar{c}$.

$$\frac{W(h - h_0)\bar{c}}{\frac{1}{2}\rho V^2 S_W \bar{c}} = \underbrace{\frac{W}{\frac{1}{2}\rho V^2 S_W}}_{C_W}(h - h_0).$$

Tail term: insert $S_T/S_T = 1$.

$$\frac{L_T l_T}{\frac{1}{2}\rho V^2 S_W \bar{c}} = \underbrace{\frac{L_T}{\frac{1}{2}\rho V^2 S_T}}_{C_L^T} \underbrace{\frac{S_T l_T}{S_W \bar{c}}}_{\bar{V}_T}.$$

Thus

$$C_M = C_{M0} + C_W(h - h_0) - C_L^T \bar{V}_T, \qquad \bar{V}_T = \frac{S_T l_T}{S_W \bar{c}}.$$

$\bar{V}_T$ is the dimensionless tailplane volume coefficient.

3. Use the level-flight force balance

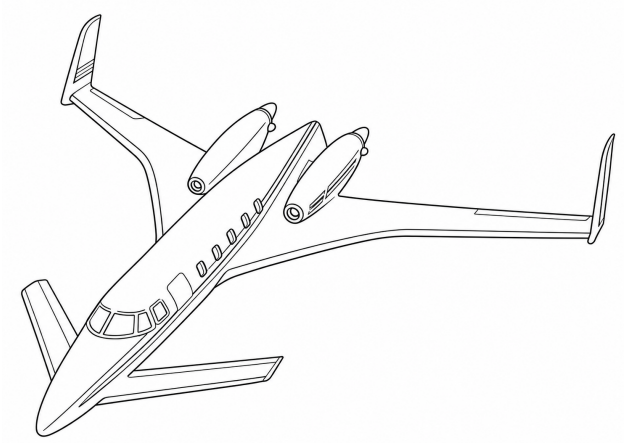
$$L = W \quad \Longrightarrow \quad \frac{L}{\frac{1}{2}\rho V^2 S_W} = \frac{W}{\frac{1}{2}\rho V^2 S_W} \quad \Longrightarrow \quad C_L = C_W.$$

$$\boxed{C_M = C_{M0} + C_L(h - h_0) - C_L^T \bar{V}_T.}$$

Pitching equilibrium requires $M = 0$, hence $C_M = 0$.

1 / STEADY LEVEL FLIGHT

Can the tailplane volume coefficient be negative?



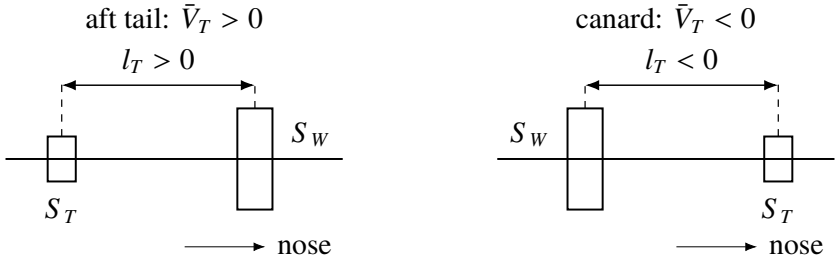
Raytheon–Beech canard configuration.

1. Keep the same meaning for the reference areas

S_W is the *main-wing* area; S_T is the aft-tail or foreplane area. Both areas and $\bar{c}$ are positive.

2. Give the longitudinal arm a sign

Measure l_T from the main-wing AC to the second surface's AC, positive aft:



3. Read the moment contribution

$$\bar{V}_T = \frac{S_T l_T}{S_W \bar{c}}, \qquad \text{second-surface moment coefficient} = -C_L^T \bar{V}_T.$$

An upward force on an aft tail gives a nose-down contribution. An upward force on a foreplane gives a nose-up contribution. The signed arm accounts for that change.

1 / STEADY LEVEL FLIGHT

Calculate tail lift for pitching equilibrium

Question. In steady level flight, an aircraft has

$$\bar{V}_T = 0.8, \quad h - h_0 = 0.2, \quad C_{M0} = -0.05, \quad C_L = 1.$$

Find the tail lift coefficient required for pitching equilibrium.

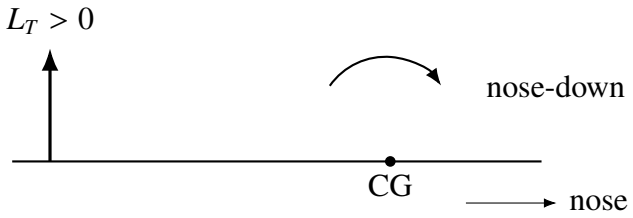
1. Solve for the unknown before inserting numbers

$$\begin{aligned} 0 &= C_{M0} + C_L(h - h_0) - C_L^T \bar{V}_T, \\ C_L^T \bar{V}_T &= C_{M0} + C_L(h - h_0), \\ C_L^T &= \frac{C_{M0} + C_L(h - h_0)}{\bar{V}_T}. \end{aligned}$$

2. Substitute and interpret the sign

$$C_L^T = \frac{-0.05 + 1 \cdot 0.2}{0.8} \simeq +0.188.$$

Moment contributions: $C_{M0} = -0.05$, $C_L(h - h_0) = +0.20$, and $-C_L^T \bar{V}_T = -0.15$.



The positive tail coefficient gives an upward tail load and a nose-down moment.

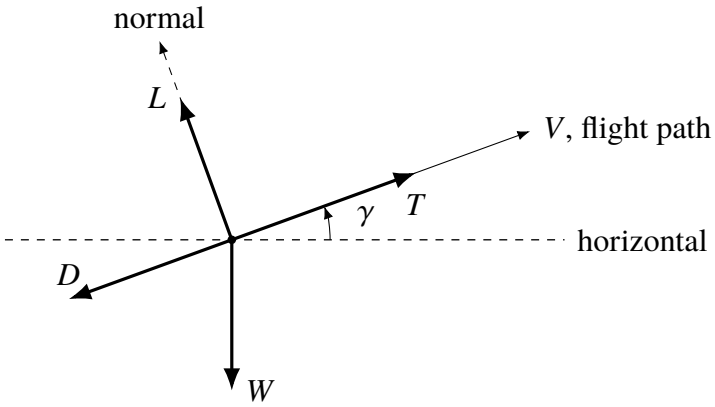
Second question. What is the weight coefficient in this level-flight condition?

$$C_W \equiv \frac{W}{\frac{1}{2}\rho V^2 S_W} = C_L = 1.$$

2 / STEADY CLIMB

Steady climb: resolve the forces

Steady climb: constant speed along a straight path at angle $\gamma > 0$ to the horizontal.



Lift is normal to V. Drag opposes V; thrust acts along V. Weight remains vertical.

1. Resolve weight in flight-path axes

$$\text{Along the path: } -W\sin\gamma, \qquad \text{Normal to the path: } -W\cos\gamma.$$

2. Apply force equilibrium

$$\begin{aligned} T - D - W\sin\gamma = 0 \quad &\Longrightarrow \quad T = D + W\sin\gamma, \\ L - W\cos\gamma = 0 \quad &\Longrightarrow \quad L = W\cos\gamma. \end{aligned}$$

3. Express the lift requirement as a coefficient

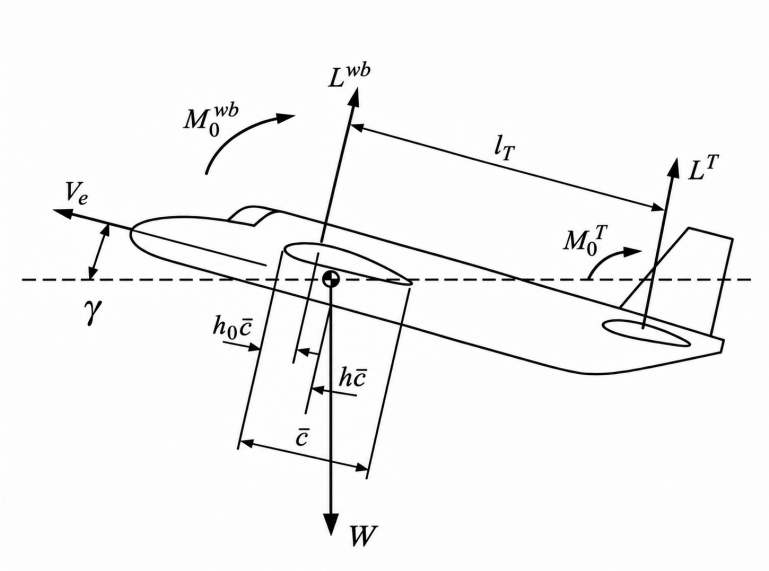
$$C_L = \frac{L}{\frac{1}{2}\rho V^2 S_W} = \frac{W\cos\gamma}{\frac{1}{2}\rho V^2 S_W}.$$

$$C_L = C_W\cos\gamma, \qquad C_W = \frac{W}{\frac{1}{2}\rho V^2 S_W}.$$

For $\gamma = 0$, these equations give $T = D$ and $L = W$.

2 / STEADY CLIMB

Steady climb: write the pitching moment



The notation and moment reference point are the same as on page 4. γ is the flight-path angle.

1. Take moments about the wing lift application point

The normal weight component is now $W\cos\gamma$:

$$M = M_0 + W\cos\gamma\,(h - h_0)\bar{c} - L_T l_T.$$

2. Divide by the same reference moment

$$\begin{aligned} C_M &= C_{M0} + \frac{W\cos\gamma\,(h-h_0)\bar{c}}{\frac{1}{2}\rho V^2 S_W \bar{c}} - \frac{L_T l_T}{\frac{1}{2}\rho V^2 S_W \bar{c}} \\ &= C_{M0} + C_W\cos\gamma\,(h-h_0) - C_L^T \bar{V}_T. \end{aligned}$$

Since $C_L = C_W\cos\gamma$,

$$\boxed{C_M = C_{M0} + C_L(h-h_0) - C_L^T \bar{V}_T.}$$

Question. How must C_L^T vary with γ to keep $C_M = 0$, if C_W is constant? Take C_{M0} and the geometry as fixed.

2 / STEADY CLIMB

Find how tail lift changes with climb angle

1. Set the climbing moment coefficient to zero

$$\begin{aligned} 0 &= C_{M0} + C_W(h - h_0)\cos\gamma - C_L^T(\gamma)\bar{V}_T, \\ C_L^T(\gamma)\bar{V}_T &= C_{M0} + C_W(h - h_0)\cos\gamma, \\ C_L^T(\gamma) &= \frac{C_{M0} + C_W(h - h_0)\cos\gamma}{\bar{V}_T}. \end{aligned}$$

2. Write the level-flight value, $\gamma = 0$

$$C_L^T(0) = \frac{C_{M0} + C_W(h - h_0)}{\bar{V}_T}.$$

3. Subtract the level-flight value

$$\begin{aligned} C_L^T(\gamma) - C_L^T(0) &= \frac{C_{M0} + C_W(h - h_0)\cos\gamma - C_{M0} - C_W(h - h_0)}{\bar{V}_T} \\ &= \frac{C_W(h - h_0)(\cos\gamma - 1)}{\bar{V}_T}. \end{aligned}$$

$$\boxed{C_L^T(\gamma) = C_L^T(0) + \frac{C_W(h - h_0)}{\bar{V}_T}(\cos\gamma - 1).}$$

Special case: $C_{M0} = 0$

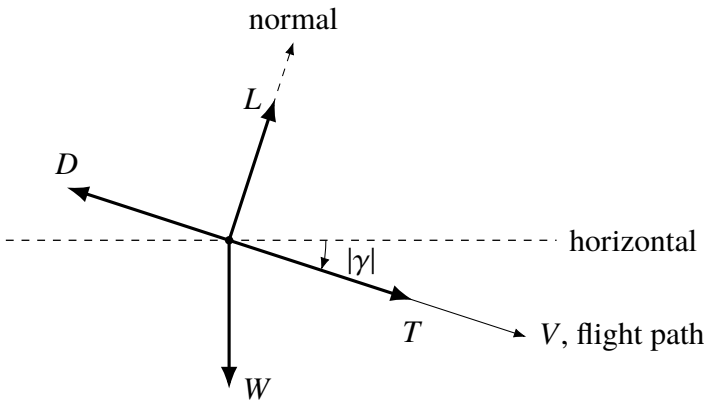
$$C_L^T(0) = \frac{C_W(h - h_0)}{\bar{V}_T} \quad \Longrightarrow \quad C_L^T(\gamma) = C_L^T(0)\cos\gamma.$$

For nonzero C_{M0}, use the general result above.

3 / STEADY DESCENT

Steady descent

Steady descent: $\gamma < 0$.



1. Use the same signed force equations

$$T - D - W\sin\gamma = 0,$$
$$L - W\cos\gamma = 0.$$

2. Use $\sin\gamma = -\sin|\gamma|$ **and** $\cos\gamma = \cos|\gamma|$

$$T = D - W\sin|\gamma|, \qquad L = W\cos|\gamma|.$$

Weight now supplies part of the force needed to overcome drag.

3. Apply pitching equilibrium

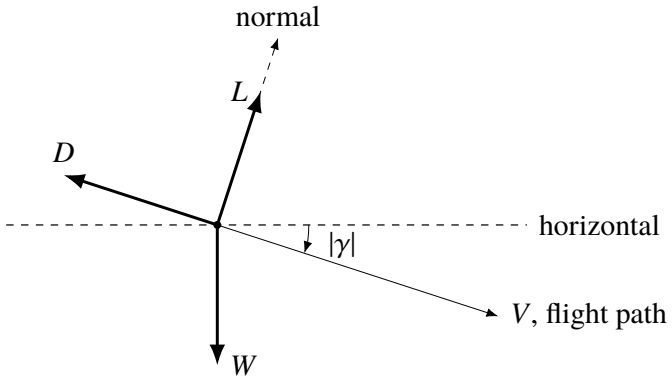
$$C_L = C_W\cos|\gamma|,$$
$$0 = C_{M0} + C_W(h - h_0)\cos|\gamma| - C_L^T\bar{V}_T,$$
$$C_L^T = \frac{C_{M0} + C_W(h - h_0)\cos|\gamma|}{\bar{V}_T}.$$

At fixed C_W, C_{M0} and geometry, equal climb and descent angle magnitudes give the same C_L and C_L^T. Their thrust requirements differ.

4 / GLIDE

Glide

A steady glide is a descent with $T = 0$.



1. Set thrust to zero in the descent equations

$$0 = D - W\sin|\gamma| \quad \Longrightarrow \quad D = W\sin|\gamma|,$$
$$L = W\cos|\gamma|.$$

2. Divide to find the glide angle

$$\frac{D}{L} = \frac{W\sin|\gamma|}{W\cos|\gamma|} = \tan|\gamma|.$$

$$\gamma = -\tan^{-1}\left(\frac{D}{L}\right).$$

A larger lift-to-drag ratio gives a shallower steady glide.

3. The aircraft must also balance in pitch

$$C_L = C_W\cos\gamma, \qquad 0 = C_{M0} + C_L(h - h_0) - C_L^T\bar{V}_T,$$
$$C_L^T = \frac{C_{M0} + C_L(h - h_0)}{\bar{V}_T}.$$

Conclusions

Flight condition	Normal force balance	Along-path balance
Level, $\gamma = 0$	$L = W$	$T = D$
Climb, $\gamma > 0$	$L = W\cos\gamma$	$T = D + W\sin\gamma$
Descent, $\gamma < 0$	$L = W\cos\lvert\gamma\rvert$	$T = D - W\sin\lvert\gamma\rvert$
Glide, $T = 0$	$L = W\cos\lvert\gamma\rvert$	$D = W\sin\lvert\gamma\rvert$

Lift coefficients

$$C_L = C_W\cos\gamma, \qquad C_W = \frac{W}{\frac{1}{2}\rho V^2 S_W},$$

$$C_L = C_L^W + \frac{S_T}{S_W}C_L^T.$$

Pitching moment and trim

$$C_M = C_{M0} + C_L(h - h_0) - C_L^T\bar{V}_T, \qquad \bar{V}_T = \frac{S_T l_T}{S_W\bar{c}}.$$

For $C_M = 0$,

$$C_L^T = \frac{C_{M0} + C_L(h - h_0)}{\bar{V}_T}.$$

Calculation order

$$\text{Flight condition} \longrightarrow (L, T) \longrightarrow C_L \longrightarrow C_L^T.$$